Nature's Algorithm Unleashed

Can Evolution Alone Master Brittle Star Locomotion?

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INTRODUCTION

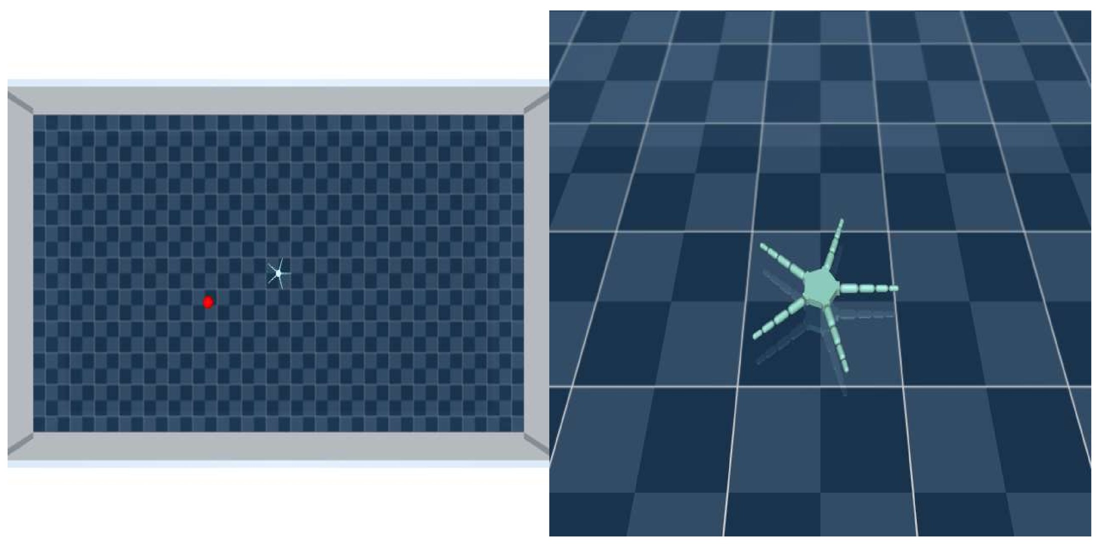
Brittle stars are remarkable marine invertebrates which use decentralized control and flexible limbs to navigate complex seabeds through **decentralized control**, achieving remarkable locomotion without central brain coordination.

This study investigates
whether evolutionary strategies **(OpenES)** with a **minimalist artificial neural network (ANN)** can replicate this efficiency, eliminating the need for higher-level Central Pattern Generators (CPGs).

We test whether evolution alone can produce:

1. **stable distance covering**
2. **energy-efficient locomotion**
3. **natural looking gait**

in robotic brittle stars, potentially advancing bio-inspired robotics through simpler, more adaptable control systems.



METHODS

We trained and optimized the parameters of the Evolutionary strategy with and without energy efficiency constraints using the same setup in 2 distinct environments.

Experimental Setup

- 2-arm morphology: 5 segments per arm (later scaled to 5 arms)
- Control system: ANN controller (no CPG) optimized via OpenES
- Training parameters: Population size = 1000, Generations = 100

Environments

- Undirected Locomotion
  - Maximize displacement from origin
  - Reward: *R1 = distance*
  - *R1* = distance / energy*
- Directed Locomotion
  - Minimize distance to target
  - Reward function: *R2 = -target_distance*
  - *R2* = -target_distance × (energy/3)*
  - Feed relative angle to target to ANN input

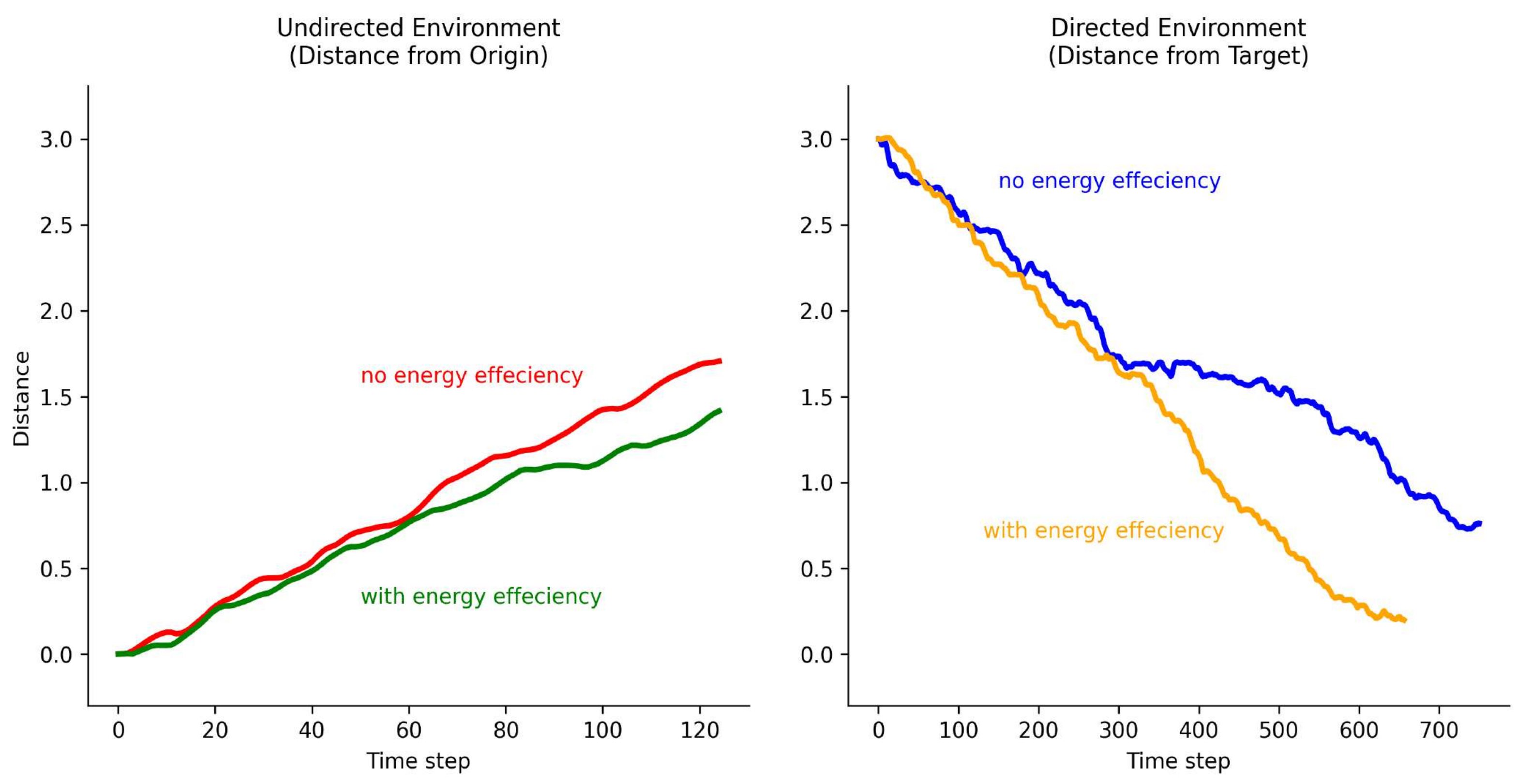
Evaluation Metrics

- Distance covered
- Energy consumption (actuator forces)
- Gait smoothness (in-plane/out-of-plane segment analysis)

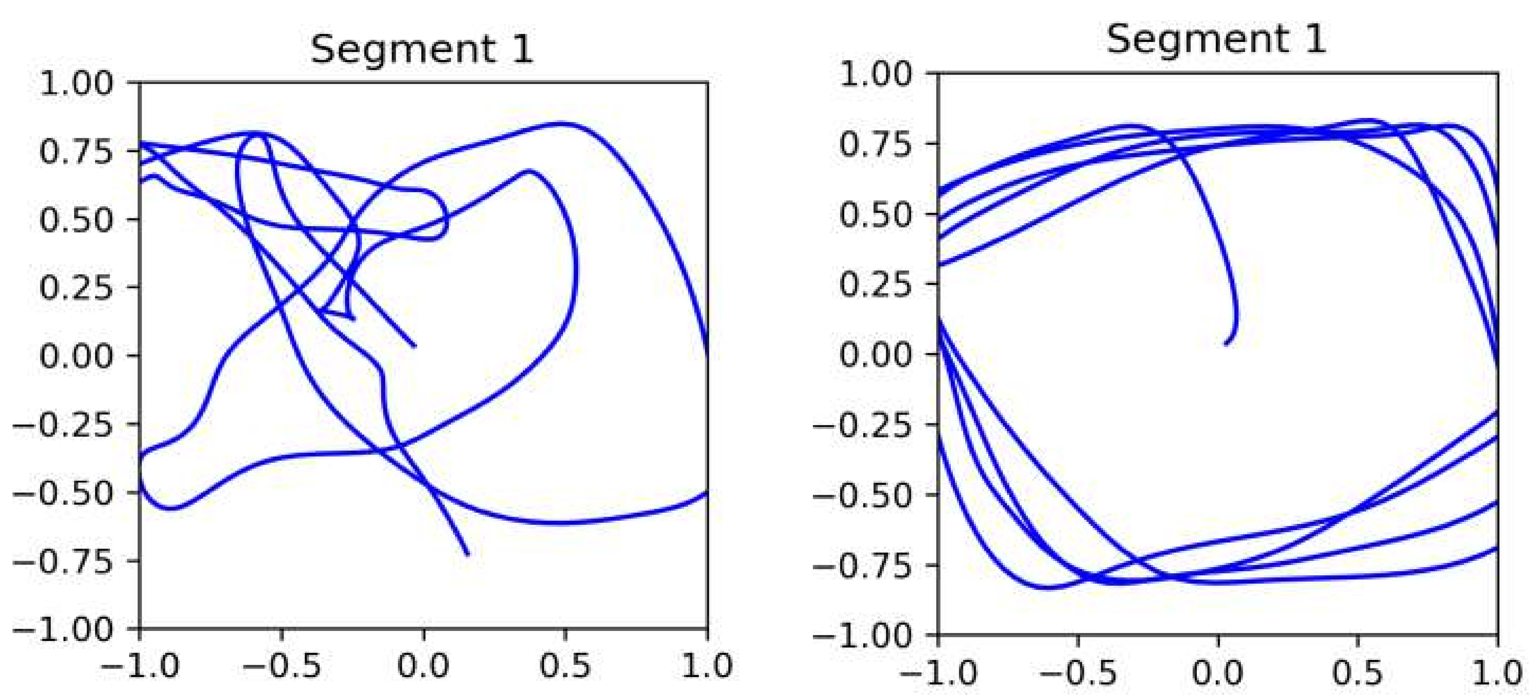
RESULTS & DISCUSSION

Key Findings for 2-Arm Configuration
Our evolutionary approach successfully generated functional locomotion in the simplified 2-arm system, demonstrating three key advantages:

- **Energy Efficiency**: Achieved 8.2% reduction in undirected and 6.0% in directed locomotion
- **Performance Trade-off**: Energy-efficient policies showed marginally lower (but still significant) displacement



- **Gait Quality**: Developed smoother, more natural movement patterns by adding Energy training constraint:
  - Reduced erratic limb motions
  - Emergent circular coordination resembling biological movement
  - Visibly more fluid locomotion in demo videos



Scaling Challenges (5-Arm Configuration)
The same approach failed to produce:

- Effective forward locomotion
- Coordinated movement across all limbs
- Stable gait patterns

CONCLUSION

Simple morphologies
OpenES with a basic ANN successfully generates energy-efficient, biologically plausible locomotion
Eliminates need for complex CPG controllers

Complex morphologies
Pure evolutionary approach shows limitations in coordination
Suggests need for hybrid control architectures

Key Implications

- Nature's advanced behavior can be replicated artificially by simple approach - but only to a point
- Morphological complexity demands architectural complexity in controllers
- Energy-efficient gaits emerge naturally when incentivized

REFERENCES

- Salimans, T., Ho, J., Chen, X., & Sutskever, I. (2017). Evolution strategies as a scalable alternative to reinforcement learning
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms

FUTURE WORK

We hope to extend our current implementation using a simple ANN controllers by setting up an intermediary CPG for more controller abstraction. As well as compare our findings to the PPO machine learning approach.

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